English: [About 30 minutes]

- **Write a story about yourself.**

Math: [About 60 minutes]

- **Please complete at least 5 questions.**
- **There are 30 questions in total. Choose at least one question from each group.**
- **Some questions have sub questions, which will be counted as extra points.**

Group 1:

1. Prove $(1 + x)^n \geq 1 + nx$ for all $x > -1$ and for all integers $n \geq 2$.

2. Describe two situations where Taylor series can be applied.

3. Find the shortest distance from the point (1, 0, −2) to the plane $x + 2y + z = 4$.

4. How do you approximate e^x and ln x? How big is the error of approximation?

5. A rectangular box with no top is to be made from 12 m^2 of cardboard. Find the maximum volume of such a box.

6. A, B, C, D, E, F, G, H and I are the nine distinct integers taking values between one to nine (not necessarily in order). They satisfy the following constraints:

$$A + B + C + D = 20$$

$$B + C + D + E + F = 20$$

$$D + E + F + G + H = 20$$

$$F + G + H + I = 20$$

Solve for A, B, C, D, E, F, G, H and I.

Group 2:

7. How do you generate two $N(0, 1)$ (standard normal) random variables with correlation p if you have a random generator for standard normal distribution?

8. Write an algorithm to carry out numerical integration. How might you optimize the algorithm?

9. Describe a root-finding algorithm to solve $f(x) = 0$ when $f(x)$ is a differentiable function. Can you provide an algorithm when $f(x)$ is continuous, but no longer differentiable everywhere? What about for any function $f(x)$?

10. How many people do we need in a class to ensure the probability that at least two people have the same birthday is more than 50%? (Assume there are 365 days a year.)

11. How many degrees are there in the angle between the hour and minute hands of a clock when the time is a quarter past three?

12. Explain the general process of least-squares linear regression.

Group 3:

13. Given the matrix

$$A = \begin{bmatrix} 5 & -3 \\ -3 & 5 \end{bmatrix}$$

a. Find eigenvalue(s) and corresponding eigenvector(s) to matrix A.

b. Find a matrix M, such that $A = M^2$.

c. Find a matrix C, such that $A = C^{\mathrm{T}}C$.

14. Consider the matrix

$$A = \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{bmatrix}$$

Find the SVD of the matrix $A = U\Sigma V^{\mathrm{T}}$.

15. For X in R^3, let $Q(X) = 5x_1^2 + 3x_2^2 + 2x_3^2 - x_1x_2 + 8x_2x_3$. Write this quadratic form as $X^{\mathrm{T}}AX$.

16. What is the rank for KK^{T}, where K is an n * m matrix?

17. Find two numbers whose sum is 14, whose average is 2, and whose difference is 4. If that is not possible, get as close as you can with the least-squares solution.

18. Give examples, compare, and contrast probability experiments involving Bernoulli, Binomial, Poisson, and Gamma distributions.

Group 4:

19. Given the table of 6 observations below:

observation	#1	#2	#3	#4	#5	#6
x	1	5	2	6	7	3
y	3	11	6	8	15	11

a. Construct the sample covariance matrix.

b. Find the principal components of the data.

20. We have a container with 300 red balls and 600 blue balls. We mix the balls well and choose one at random, with each ball being equally likely to be chosen. After each choice, we return the chosen ball to the container and mix again.

a. What is the probability that the first n balls chosen are all blue?

b. Let N be the number of blue balls chosen before the first red one. What is the P (N = n)? What are the mean and variance of N? Explain your answers using the formulae

$$\sum_{n=0}^{\infty} x^n = \frac{1}{1-x} \quad \text{for } |x| < 1$$

$$\sum_{n=0}^{\infty} nx^n = x\frac{d}{dx}\frac{1}{1-x} \quad \text{for } |x| < 1$$

etc.

c. What is the probability that N = 0 given that $N \leq 2$?

d. What is the probability that N is an even number?

21. A tourist decides between two plays, called “Good” (G) and “Bad” (B). The probability of the tourist choosing Good is $P(G)$ = 0.1. In addition, a tourist either “Likes” (L) the play or “dislikes” (D) the play. The probability the tourist likes a good play is P ($L \mid G$) = 0.7 while disliking a bad play is P ($D \mid B$) = 0.8.

a. Draw a probability decision tree diagram to illustrate the choices.

b. Calculate and interpret $P(L)$.

c. If the tourist liked the play they chose, what is the probability they chose “Good”?

22. A “triangular” random variable, X has probability density function (pdf)

$$f(x) = \begin{cases} 2(1-x) & \text{if } 0 \leq x \leq 1, \\ 0 & \text{otherwise.} \end{cases}$$

a. Calculate the mean and variance of X.

b. Suppose X_1 and X_2 are independent and identically distributed (i.i.d) copies of X and $Y = X_1 + X_2$ Find the pdf for Y.

c. Calculate the mean and variance of Y without using the pdf found above.

d. Find P ($Y > 1$).

e. Suppose X_1, X_2, ..., X_{100} are i.i.d copies of X. Estimate $P(X_1 + \ldots + X_{100} > 34)$ using the central limit theorem.

23. Suppose X and Y have a joint pdf

$$f(x,y) = \frac{1}{8\pi} \begin{cases} 4 - x^2 - y^2 & \text{if } x^2 + y^2 \leq 4, \\ 0 & \text{otherwise.} \end{cases}$$

a. Calculate $P(X_2 + Y_2 \leq 1)$

b. Calculate the marginal pdf of X.

c. What is the covariance of X and Y?

d. Find an event depending on only X whose probability depends on Y. Use this to show that X and Y are not independent.

e. Write the joint PDF for $U = X^2$ and $V = Y^2$.

f. Calculate the covariance between X^2 and Y^2. It may be simpler to do this independent of the above part. Use this to show X and Y are not independent.

24. You have a fair coin and you want to simulate an event that has probability 1/3, and another event that has probability 2/3. How would you go about simulating these events?

Group 5

25. Give an example of a distribution with infinite variance.

26. We play a game: I pick a number n from 1 to 100. If you guess correctly, I pay you $n and zero otherwise. How much would you pay to play this game?

27. A company is holding a dinner for working members with at least one son. Mrs. Johnson, a mother with two children, is invited. What is the probability that both children are boys?

28. What is a positive-definite matrix? What properties does a positive-definite matrix have?

29. If X is $N(\mu, \sigma)$ and $\lambda > 0$, calculate the values of $E(X^2)$ and $E(e^{\lambda X})$.

30. The tables below display the multiple regression coefficient estimates when TV, radio, and newspaper advertising budgets are used to predict product sales. In addition, extra information about the model is provided.

	Coefficient	Std. error	t-statistic	p-value
Intercept	2.939	0.3119	9.42	< 0.0001
TV	0.046	0.0014	32.81	< 0.0001
radio	0.189	0.0086	21.89	< 0.0001
newspaper	−0.001	0.0059	−0.18	0.8599

Quantity	Value
Residual standard error	1.69
R^2	0.897
F-statistic	570

a. Using the first table, write down the regression model for the response variable sales.

b. Interpret the meaning of coefficient “0.189” in the above model.

c. In the first table, column 4 and row 3, the t-statistic equals 32.81. What is the meaning of this, and what is the underlying test? Write out the null and alternative hypothesis for this test. What conclusion can you draw?

d. Estimate the confidence interval for the coefficient 0.189 for radio.

e. Interpret the meaning of R^2 = 0.897.

f. How well does the model fit the data? Why do you think this is the case?

g. List assumptions for the classic multiple linear regression model.

h. Assume you are given the table below, which contains the correlation matrix for TV, radio, newspaper, and sales. Is your model in the first part still okay? Why do you think so? If so, how would you address this problem?

	TV	radio	newspaper	sales
TV	1.0000	0.0548	0.0567	0.7822
radio		1.0000	0.3541	0.5762
newspaper			1.0000	0.2283
sales				1.0000